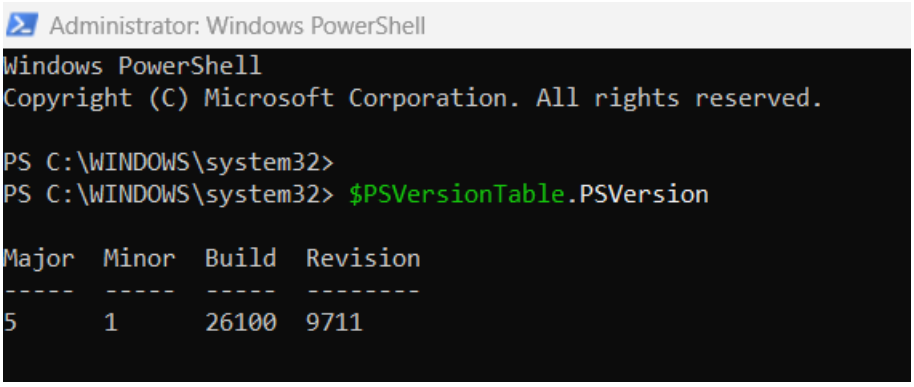
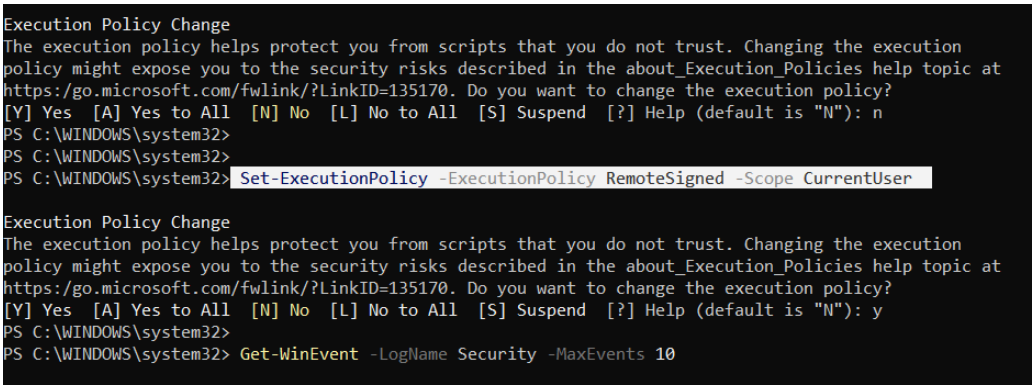
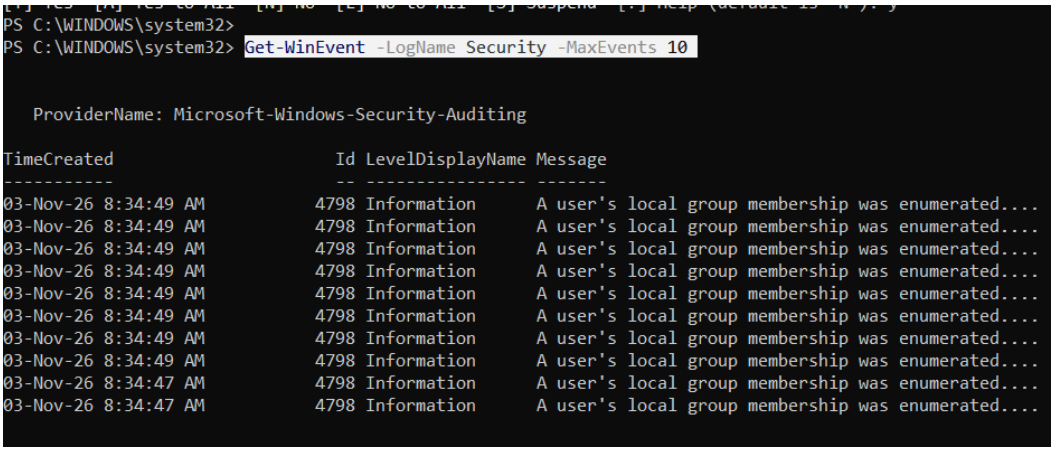
Week – 4 - Windows-Log-Analysis-with-PowerShell



Exercise 1: View Recent Security Events





Exercise 2: Filter by Event ID

```
User@ROKON-LENEVO PS C:\WINDOWS\system32> Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4624} -MaxEvents 20


   ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
03-Nov-26 2:42:04 PM          4624 Information      An account was successfully logged on....
03-Nov-26 2:38:32 PM          4624 Information      An account was successfully logged on....
03-Nov-26 2:38:30 PM          4624 Information      An account was successfully logged on....
03-Nov-26 2:38:30 PM          4624 Information      An account was successfully logged on....
03-Nov-26 8:37:18 AM          4624 Information      An account was successfully logged on....
03-Nov-26 8:37:18 AM          4624 Information      An account was successfully logged on....
03-Nov-26 8:37:12 AM          4624 Information      An account was successfully logged on....
03-Nov-26 8:28:23 AM          4624 Information      An account was successfully logged on....
03-Nov-26 8:27:15 AM          4624 Information      An account was successfully logged on....
03-Nov-26 8:26:59 AM          4624 Information      An account was successfully logged on....
03-Nov-26 8:23:04 AM          4624 Information      An account was successfully logged on....
03-Nov-26 8:23:04 AM          4624 Information      An account was successfully logged on....
03-Nov-26 8:22:44 AM          4624 Information      An account was successfully logged on....
03-Nov-26 7:50:54 AM          4624 Information      An account was successfully logged on....
03-Nov-26 7:50:52 AM          4624 Information      An account was successfully logged on....
03-Nov-26 7:50:52 AM          4624 Information      An account was successfully logged on....
03-Nov-26 7:35:36 AM          4624 Information      An account was successfully logged on....
03-Nov-26 7:35:27 AM          4624 Information      An account was successfully logged on....
03-Nov-26 7:32:48 AM          4624 Information      An account was successfully logged on....
03-Nov-26 7:20:41 AM          4624 Information      An account was successfully logged on....
```

```
User@ROKON-LENEVO PS C:\WINDOWS\system32>
User@ROKON-LENEVO PS C:\WINDOWS\system32> Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4624} -MaxEvents 20
>>     Select-Object TimeCreated, Id, Message |
>>     Format-Table -AutoSize

TimeCreated            Id Message
-----------            -- -------
03-Nov-26 2:42:04 PM 4624 An account was successfully logged on....
03-Nov-26 2:38:32 PM 4624 An account was successfully logged on....
03-Nov-26 2:38:30 PM 4624 An account was successfully logged on....
03-Nov-26 2:38:30 PM 4624 An account was successfully logged on....
03-Nov-26 8:37:18 AM 4624 An account was successfully logged on....
03-Nov-26 8:37:18 AM 4624 An account was successfully logged on....
03-Nov-26 8:37:12 AM 4624 An account was successfully logged on....
03-Nov-26 8:28:23 AM 4624 An account was successfully logged on....
03-Nov-26 8:27:15 AM 4624 An account was successfully logged on....
03-Nov-26 8:26:59 AM 4624 An account was successfully logged on....
03-Nov-26 8:23:04 AM 4624 An account was successfully logged on....
03-Nov-26 8:23:04 AM 4624 An account was successfully logged on....
03-Nov-26 8:22:44 AM 4624 An account was successfully logged on....
03-Nov-26 7:50:54 AM 4624 An account was successfully logged on....
03-Nov-26 7:50:52 AM 4624 An account was successfully logged on....
03-Nov-26 7:50:52 AM 4624 An account was successfully logged on....
03-Nov-26 7:35:36 AM 4624 An account was successfully logged on....
03-Nov-26 7:35:27 AM 4624 An account was successfully logged on....
03-Nov-26 7:32:48 AM 4624 An account was successfully logged on....
03-Nov-26 7:20:41 AM 4624 An account was successfully logged on....
```

Exercise 3: Filter by Time Range

```
User@ROKON-LENEVO PS C:\WINDOWS\system32>
User@ROKON-LENEVO PS C:\WINDOWS\system32> $StartTime = (Get-Date).AddHours(-1)
>> Get-WinEvent -FilterHashtable @{
>>     LogName='Security'
>>     StartTime=$StartTime
>> }


   ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
03-Nov-26 2:57:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:57:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:57:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:57:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:57:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:57:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:57:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:57:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:57:17 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:57:17 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:57:17 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:57:17 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:57:17 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:57:17 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:57:17 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:56:50 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:56:50 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:56:50 PM          4798 Information      A user's local group membership was enumerated....
```

```
>> } -MaxEvents 100$StartTime = (Get-Date).AddDays(-1) (Get-Date).AddDays(-1)
>> Get-WinEvent -FilterHashtable @{t was successfully logged on....
>>     LogName='Security'
>>     StartTime=$StartTime
>> } -MaxEvents 100S C:\WINDOWS\system32>
Get-WinEvent : Cannot bind parameter 'MaxEvents'. Cannot convert value "10011/02/2026 14:55:57" to type "System.I
At line:5 char:14
+ } -MaxEvents 100$StartTime = (Get-Date).AddDays(-1)
+              ~~~~~~~~~~~~~~~
    + CategoryInfo          : InvalidArgument: (:) [Get-WinEvent], ParameterBindingException
    + FullyQualifiedErrorId : CannotConvertArgumentNoMessage,Microsoft.PowerShell.Commands.GetWinEventCommand



   ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
03-Nov-26 2:55:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:55:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:55:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:55:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:55:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:55:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:55:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:55:44 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:55:17 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:55:17 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:55:17 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:55:17 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:55:17 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:55:17 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:55:17 PM          4798 Information      A user's local group membership was enumerated....
03-Nov-26 2:54:56 PM          4798 Information      A user's local group membership was enumerated....
```

```
User@ROKON-LENEVO PS C:\WINDOWS\system32> $StartTime = Get-Date "2024-01-01 00:00:00"
>> $EndTime = Get-Date "2024-01-02 00:00:00"
>> Get-WinEvent -FilterHashtable @{
>>     LogName='Security'
>>     StartTime=$StartTime
>>     EndTime=$EndTime
>> }
Get-WinEvent : No events were found that match the specified selection criteria.
At line:3 char:1
+ Get-WinEvent -FilterHashtable @{
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : ObjectNotFound: (:) [Get-WinEvent], Exception
    + FullyQualifiedErrorId : NoMatchingEventsFound,Microsoft.PowerShell.Commands.GetWinEventCommand

User@ROKON-LENEVO PS C:\WINDOWS\system32>
```

Exercise 4: Count Events

```
User@ROKON-LENEVO PS C:\WINDOWS\system32> $StartTime = (Get-Date).AddDays(-1)
>> $FailedLogons = Get-WinEvent -FilterHashtable @{
>>     LogName='Security'
>>     ID=4625
>>     StartTime=$StartTime
>> }
>> $FailedLogons.Count
5
```

Part 3: Extracting Event Properties (60 minutes)

Exercise 5: Understanding Event Structure

```
User@ROKON-LENEVO PS C:\WINDOWS\system32> $Event = Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625} -MaxEvents 1
>> $Event | Format-List *


Message             : An account failed to log on.

                      Subject:
                       Security ID:           S-1-5-18
                       Account Name:          ROKON-LENEVO$
                       Account Domain:        WORKGROUP
                       Logon ID:              0x3E7

                      Logon Type:                     2

                      Account For Which Logon Failed:
                       Security ID:           S-1-0-0
                       Account Name:          User
                       Account Domain:        ROKON-LENEVO

                      Failure Information:
                       Failure Reason:        Unknown user name or bad password.
                       Status:                0xC000006D
                       Sub Status:            0xC000006A

                      Process Information:
                       Caller Process ID:     0xb38
                       Caller Process Name:   C:\Windows\System32\svchost.exe

                      Network Information:
                       Workstation Name:      ROKON-LENEVO
                       Source Network Address: 127.0.0.1
                       Source Port:           0

                      Detailed Authentication Information:
                       Logon Process:         User32
                       Authentication Package: Negotiate
                       Transited Services:    -
                       Package Name (NTLM only):       -
                       Key Length:            0
```

```
Id                     : 4625
Version                : 0
Qualifiers             :
Level                  : 0
Task                   : 12544
Opcode                 : 0
Keywords               : -9218868437227405312
RecordId               : 3422623
ProviderName           : Microsoft-Windows-Security-Auditing
ProviderId             : 54849625-5478-4994-a5ba-3e3b0328c30d
LogName                : Security
ProcessId              : 1484
ThreadId               : 11812
MachineName            : ROKON-Lenevo
UserId                 :
TimeCreated            : 03-Nov-26 8:23:01 AM
ActivityId             : dac940b0-6b2d-0004-a042-c9da2d6bdd01
RelatedActivityId      :
ContainerLog           : Security
MatchedQueryIds        : {0}
Bookmark               : System.Diagnostics.Eventing.Reader.EventBookmark
LevelDisplayName       : Information
OpcodeDisplayName      : Info
TaskDisplayName        : Logon
KeywordsDisplayNames   : {Audit Failure}
```

Exercise 6: Extract Data from Failed Logons

```
User@ROKON-LENEVO PS C:\WINDOWS\system32> Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625} -MaxEvents 10
>>     ForEach-Object {
>>         [PSCustomObject]@{
>>             TimeCreated = $_.TimeCreated
>>             TargetUser = $_.Properties[5].Value
>>             SourceIP = $_.Properties[19].Value
>>             LogonType = $_.Properties[8].Value
>>             FailureReason = $_.Properties[13].Value
>>         }
>>     } | Format-Table -AutoSize

TimeCreated           TargetUser SourceIP  LogonType FailureReason
-----------           ---------- --------  --------- -------------
03-Nov-26 8:23:01 AM  User       127.0.0.1 %%2313    ROKON-LENEVO
03-Nov-26 8:22:56 AM  User       127.0.0.1 %%2313    ROKON-LENEVO
03-Nov-26 8:22:52 AM  User       127.0.0.1 %%2313    ROKON-LENEVO
03-Nov-26 8:35:48 AM  User       127.0.0.1 %%2313    ROKON-LENEVO
02-Nov-26 10:11:28 PM User       127.0.0.1 %%2313    ROKON-LENEVO
01-Nov-26 3:43:24 PM  User       127.0.0.1 %%2313    ROKON-LENEVO
01-Nov-26 2:26:16 PM  User       127.0.0.1 %%2313    ROKON-LENEVO
```

Exercise 7: Analyze Failed Logon Attempts

```
User@ROKON-LENEVO PS C:\WINDOWS\system32> Get-WinEvent -FilterHashtable @{
>>     LogName='Security'
>>     ID=4625
>>     StartTime=(Get-Date).AddDays(-1)
>> } | ForEach-Object {
>>     $_.Properties[19].Value
>> } | Where-Object { $_ -ne '-' -and $_ -ne '' } |
>>     Group-Object |
>>     Sort-Object Count -Descending |
>>     Select-Object -First 5 Name, Count

Name       Count
----       -----
127.0.0.1      5
```

```
User@ROKON-LENEVO PS C:\WINDOWS\system32> Get-WinEvent -FilterHashtable @{
>>     LogName='Security'
>>     ID=4625
>>     StartTime=(Get-Date).AddDays(-1)
>> } | ForEach-Object {
>>     $_.Properties[19].Value
>> } | Where-Object { $_ -ne '-' -and $_ -ne '' } |
>>     Group-Object |
>>     Sort-Object Count -Descending |
>>     Select-Object -First 5 @{Name='IP Address';Expression={$_.Name}}, @{Name='Failed Attempts';Expression={$_.Count}} |
>>     Format-Table -AutoSize

IP Address Failed Attempts
---------- --------------
127.0.0.1               5
```

Exercise 8: Identify Targeted Accounts

```
User@ROKON-LENEVO PS C:\WINDOWS\system32> Get-WinEvent -FilterHashtable @{
>>     LogName='Security'
>>     ID=4625
>>     StartTime=(Get-Date).AddDays(-1)
>> } | ForEach-Object {
>>     $_.Properties[5].Value  # TargetUserName
>> } | Where-Object { $_ -ne '-' -and $_ -ne '' } |
>>     Group-Object |
>>     Sort-Object Count -Descending |
>>     Select-Object -First 5 @{Name='Username';Expression={$_.Name}}, @{Name='Failed Attempts';Expression={$_.Count}} |
>>     Format-Table -AutoSize

Username Failed Attempts
-------- ---------------
User                   5
```

Exercise 9: Analyze Logon Types

```
User@ROKON-LENEVO PS C:\WINDOWS\system32> Get-WinEvent -FilterHashtable @{
>>     LogName='Security'
>>     ID=4625
>>     StartTime=(Get-Date).AddDays(-1)
>> } | ForEach-Object {
>>     $LogonType = $_.Properties[8].Value
>>     switch ($LogonType) {
>>         2 { 'Interactive' }
>>         3 { 'Network' }
>>         4 { 'Batch' }
>>         5 { 'Service' }
>>         7 { 'Unlock' }
>>         8 { 'NetworkCleartext' }
>>         9 { 'NewCredentials' }
>>         10 { 'RemoteInteractive' }
>>         11 { 'CachedInteractive' }
>>         default { "Unknown ($LogonType)" }
>>     }
>> } | Group-Object |
>>     Sort-Object Count -Descending |
>>     Select-Object @{Name='Logon Type';Expression={$_.Name}}, @{Name='Count';Expression={$_.Count}}
>>     Format-Table -AutoSize

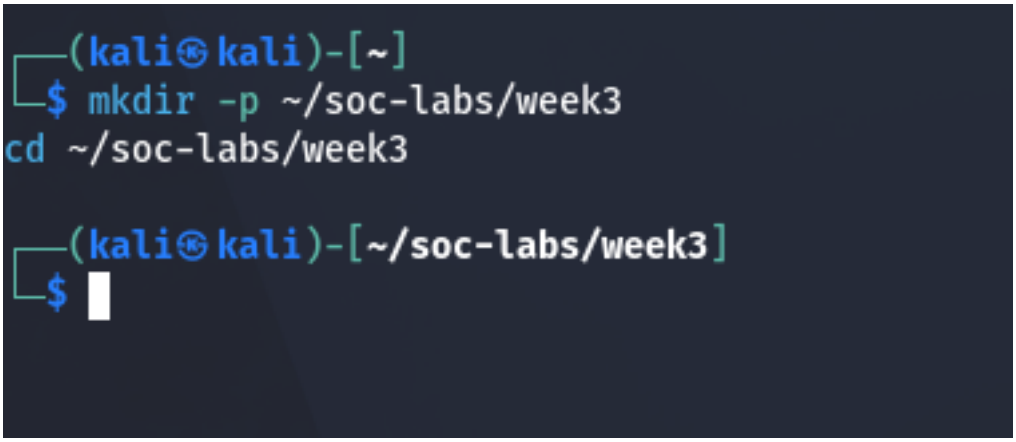
Logon Type       Count
----------       -----
Unknown (%%2313)     5
```

Week – 3 Lab-Log-Analysis-with-CLI-Tools

Create a working directory:

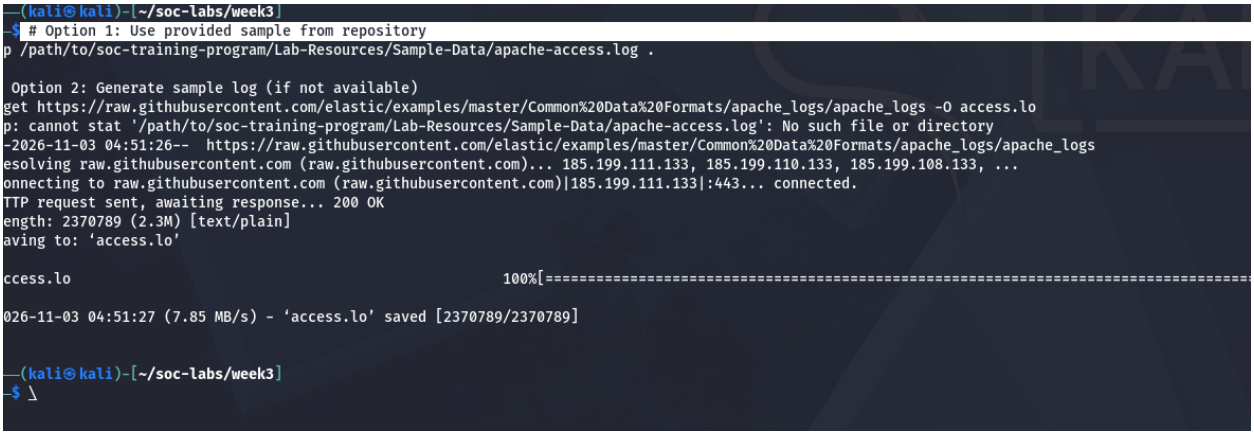
mkdir -p ~/soc-labs/week3

cd ~/soc-labs/week3



Download sample log file:

wget https://raw.githubusercontent.com/elastic/examples/master/Common%20Data%20Formats/apache_logs/apache_logs -O access.log



Verify the file:

ls -lh access.log

head -5 access.log

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ ls -lh access.log
head -5 access.log
-rw-rw-r-- 1 kali kali 2.3M Nov  3 04:54 access.log
83.149.9.216 - - [17/May/2015:10:05:03 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-search.png HTTP/1.1" 200 203023 "http://semicomplete.com/presentations/logstas
OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
83.149.9.216 - - [17/May/2015:10:05:43 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-dashboard3.png HTTP/1.1" 200 171717 "http://semicomplete.com/presentations/log
Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
83.149.9.216 - - [17/May/2015:10:05:47 +0000] "GET /presentations/logstash-monitorama-2013/plugin/highlight/highlight.js HTTP/1.1" 200 26185 "http://semicomplete.com/presentations/log
Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
83.149.9.216 - - [17/May/2015:10:05:12 +0000] "GET /presentations/logstash-monitorama-2013/plugin/zoom-js/zoom.js HTTP/1.1" 200 7697 "http://semicomplete.com/presentations/logstash-mo
 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
83.149.9.216 - - [17/May/2015:10:05:07 +0000] "GET /presentations/logstash-monitorama-2013/plugin/notes/notes.js HTTP/1.1" 200 2892 "http://semicomplete.com/presentations/logstash-mon
10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$
```

Exercise 1: Count Total Requests

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ wc -l access.log
10000 access.log

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$
```

Exercise 2: Extract IP Addresses

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ awk '{print $1}' access.log | head -10
83.149.9.216
83.149.9.216
83.149.9.216
83.149.9.216
83.149.9.216
83.149.9.216
83.149.9.216
83.149.9.216
83.149.9.216
83.149.9.216

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$
```

Exercise 3: Find Top Talkers

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ awk '{print $1}' access.log | sort | uniq -c | sort -rn | head -10
    482 66.249.73.135
    364 46.105.14.53
    357 130.237.218.86
    273 75.97.9.59
    113 50.16.19.13
    102 209.85.238.199
     99 68.180.224.225
     84 100.43.83.137
     83 208.115.111.72
     82 198.46.149.143

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$
```

Exercise 4: Analyze HTTP Status Codes

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ grep " 404 " access.log | wc -l
213

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$
```

Exercise 5: Identify Suspicious URLs

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ awk '$9 == 404 {print $7}' access.log | sort | uniq -c | sort -rn | head -5
     61 /files/logstash/logstash-1.3.2-monolithic.jar
     32 /presentations/logstash-puppetconf-2012/images/office-space-printer-beat-down-gif.gif
      6 /wp/wp-admin/
      6 /wp-login.php?action=register
      6 /wp-login.php

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$
```

Exercise 6: Detect Scanning Activity

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ grep -i "nikto" access.log | wc -l
0

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$
```

Exercise 7: Detect SQL Injection Attempts

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ grep -iE "(union|select|insert|update|delete|drop|exec|script)" access.log | wc -l
124

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$
```

Exercise 8: Detect XSS Attempts

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ grep -iE "(<script|javascript:|onerror=|%3Cscript)" access.log

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ grep -iE "(<script|javascript:|onerror=|%3Cscript)" access.log | wc -l
0

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$
```

Exercise 9: Detect Directory Traversal

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ grep -E "(\.\./|\.\.%2[Ff])" access.log

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ grep -E "(\.\./|\.\.%2[Ff])" access.log | wc -l
0

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$
```

Exercise 10: Analyze Traffic by Time



Exercise 11: Analyze Request Methods



Exercise 12: Extract POST Requests



Exercise 13: Calculate Bandwidth Usage

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ awk '{sum += $10} END {print sum/1024/1024 " MB"}' access.log
2620.01 MB

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$
```

Exercise 14: Extract and Format IPs

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ awk '{print $1}' access.log | sort -u > unique_ips.txt

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ wc -l unique_ips.txt
ead unique_ips.txt
753 unique_ips.txt
00.2.4.116
00.43.83.137
01.119.18.35
01.199.108.50
01.226.168.196
01.226.168.198
01.226.33.222
03.245.44.13
03.247.192.5
03.25.13.22
```

Exercise 15: Anonymize IP Addresses

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ sed 's/\([0-9]\{1,3\}\.\)\{3\}[0-9]\{1,3\}/\1XXX/' access.log | head -5
.XXX - - [17/May/2015:10:05:03 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-search.png HTTP/1.1" 200 203023
9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
.XXX - - [17/May/2015:10:05:43 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-dashboard3.png HTTP/1.1" 200 171
 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
.XXX - - [17/May/2015:10:05:47 +0000] "GET /presentations/logstash-monitorama-2013/plugin/highlight/highlight.js HTTP/1.1" 200 26
 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
.XXX - - [17/May/2015:10:05:12 +0000] "GET /presentations/logstash-monitorama-2013/plugin/zoom-js/zoom.js HTTP/1.1" 200 7697 "htt
 AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
.XXX - - [17/May/2015:10:05:07 +0000] "GET /presentations/logstash-monitorama-2013/plugin/notes/notes.js HTTP/1.1" 200 2892 "http
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
```

Exercise 16: Extract URLs Only

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ awk '{print $7}' access.log | sort -u > urls.txt

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ awk '{print $7}' access.log | grep -E "\.(php|asp|jsp)$" | sort -u
/administrator/index.php
/admin.php
/wp-login.php
```

Step 17: Document Your Findings

```
┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ awk '{print $7}' access.log | grep -E "\.(php|asp|jsp)$" | sort -u
/administrator/index.php
/admin.php
/wp-login.php

┌──(kali㉿kali)-[~/soc-labs/week3]
└─$ # Apache Log Analysis Report

**Analyst:** [Your Name]
**Date:** [Analysis Date]
**Log File:** access.log
**Analysis Period:** [Date range from logs]

---

## Executive Summary

[2-3 sentence summary of key findings]

---

## 1. Traffic Overview

### Total Requests
**Answer:** [number]
```